21-242 Recitation 12

Dimension — Q3. Injective maps preserve dimension of the image

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Show that if $T : V \to W$ is an injective linear transformation, $\dim(\text{im}(T)) = \dim(V)$.

Course-level solution

Assume $\dim V = n < \infty$ (otherwise both sides are infinite if T is injective and V is infinite-dimensional, with the usual caveats). Let $(\vec{v}_1, \ldots, \vec{v}_n)$ be a basis of V. Linear Transformations Q6 says $\big(T(\vec{v}_1), \ldots, T(\vec{v}_n)\big)$ is independent; Linear Transformations Q5 says it spans $\text{im}(T)$. So it is a basis of $\text{im}(T)$ of length n, and $\dim \text{im}(T) = n = \dim V$.
(Equivalently: $T : V \to \text{im}(T)$ is an isomorphism, and isomorphisms preserve the length of bases.)

Later-course solution (using ideas from the rest of the handout)

Rank-nullity: $\dim \text{im}(T) = \dim V - \dim \ker T = \dim V$ because $\ker T = \{\vec{0}\}$.